Production Networks under Real Exchange Rate Uncertainty

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Introduction

Motivation

- Countries differ in volatility of **real exchange rate (RER)**

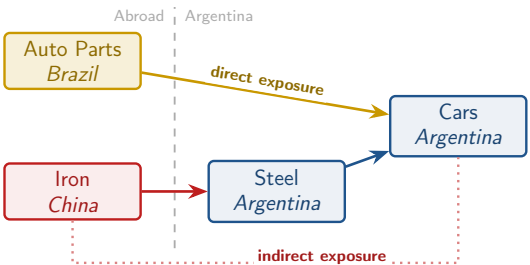
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- Movements in RER affect prices and consumption
  - **Channel:** direct and indirect import content embedded in the supply chain

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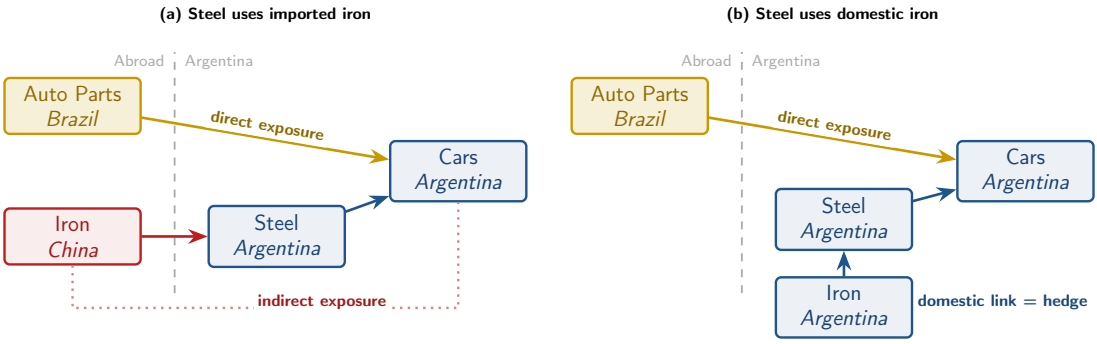
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(a) Steel uses imported iron

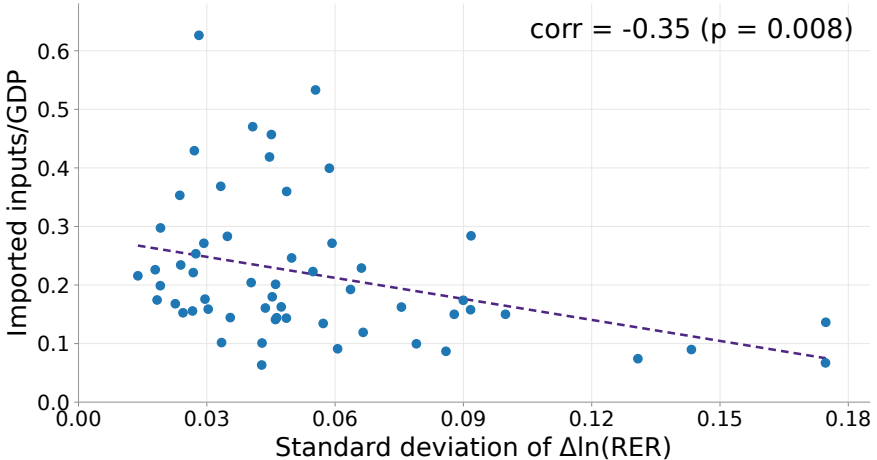


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- Movements in RER affect prices and consumption
  - **Channel:** direct and indirect import content embedded in the supply chain



Empirical motivation: Higher RER volatility, lower import exposure



58 countries, 1996–2022. Imported inputs: direct + indirect content (OECD ICIO); RER: BIS.

▸ Sector import exposure

Research question

How does exchange rate uncertainty affect firms' sourcing decisions and the resulting production structure?

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- What are the consequences for **GDP**?
- How would different **exchange rate regimes** affect the network?
- How do shocks **propagate** and **shape** the network?
- How does RER volatility impact countries' **development** and **industrialization**?

What I do

Model

- Small open economy with an **endogenous production network**
- Exogenous **exchange rate volatility**
- Firms choose suppliers **before** knowing the exchange rate

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Contribution

- Extend endogenous networks under uncertainty (Kopytov, Mishra, Nimark & Taschereau-Dumouchel 2024) to an open economy
- Study exchange rate propagation with an endogenous network

▸ Related literature

Model

Environment

Agents

- **Household:** supplies labour, consumes, owns firms.
- **Firms:** n competitive sectors and one exporter.

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Shocks

$$(z_1, \ldots, z_n,\ z_X,\ \tau,\ s) \sim \mathcal{N}(\mu, \Sigma)$$

z_i	log productivity, sector i
z_X	log productivity, exporter
$p_X^*,\ p^*$	log export, import price
$\tau \equiv p_X^* - p^*$	log terms of trade
$s = \ln S$	log nominal exchange rate (domestic per foreign)

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Technology: normalized Cobb–Douglas

$$Q_i(\alpha_i, X_i, L_i) = e^{z_i} A_i(\alpha_i) \left(\frac{L_i}{\alpha_{iL}}\right)^{\alpha_{iL}} \prod_{j:\alpha_{ij}>0} \left(\frac{X_{ij}}{\alpha_{ij}}\right)^{\alpha_{ij}} \left(\frac{X_{i*}}{\alpha_{i*}}\right)^{\alpha_{i*}}$$

X_{ij}: domestic input from j X_{i*}: imported input α: input shares $A_i(\alpha_i)$: technology shifter

Zero-share inputs are omitted.

▸ Details

Timing



Key choice: firms choose the shares α_i **before** shocks are realized

$$\alpha_i = (\alpha_{i1}, \ldots, \alpha_{in}, \alpha_{i*}), \qquad \alpha_{iL} = 1 - \textstyle\sum_j \alpha_{ij} - \alpha_{i*}$$

▸ Unit-cost derivation

Ex-ante sourcing

Firms choose α_i to maximize expected discounted profits:

$$\alpha_i^\star \in \arg\max_{\alpha_i \in \mathcal{A}_i} \mathbb{E}\Big[\underbrace{\Lambda}_{\text{SDF}} Q_i \left(P_i - \underbrace{\mathrm{UC}_i(\alpha_i, P)}_{\text{unit cost}}\right)\Big]$$

SDF: value of one more unit of consumption in each state

Lemma

In equilibrium, the shares of firm i satisfy the first-order conditions of

$$\min_{\alpha_i \in \mathcal{A}_i} \mathbb{E}\big[\mathrm{uc}_i(\alpha_i)\big] + \mathrm{Cov}\left[\lambda, \mathrm{uc}_i(\alpha_i)\right], \qquad \lambda = \ln \Lambda, \quad \mathrm{uc}_i = \ln \mathrm{UC}_i$$

Firms prefer α_i with low expected unit cost and low unit cost when marginal utility is high

▸ Derivation

Optimal import sourcing

$$\overbrace{\frac{\partial a_i}{\partial \alpha_{i*}}}^{\text{Marginal productivity}} = \overbrace{\underbrace{\mathbb{E}[s + p^* - w]}_{\text{Expected relative price}} + \underbrace{\text{Cov}(\lambda, s + p^* - w)}_{\text{Risk}}}^{\text{Marginal cost}}$$

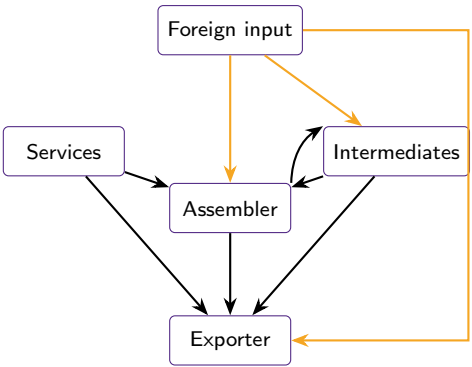
If **depreciations** coincide with bad states of the economy, imported inputs carry an extra cost.

α_{i*}: share of imported inputs $\quad a_i = \ln A_i$: technology shifter

▸ Details ▸ Correlation and volatility

Preliminary results

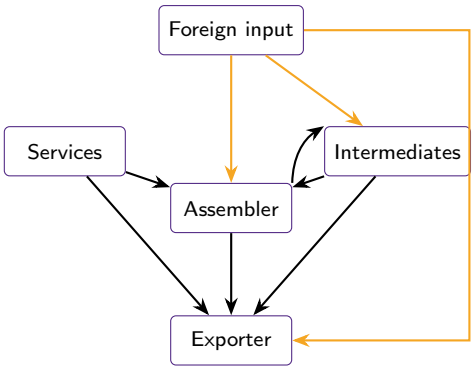
Experiment: Three sectors and exporter



Supplier → buyer; orange arrows denote imports.

▸ Experiment details ▸ Parametrization ▸ RER volatility

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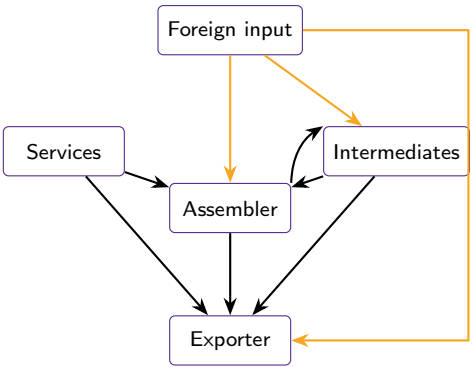


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- Raise import-price risk: sd(p^*) from 0 to 30%

Experiment details | Parametrization | RER volatility

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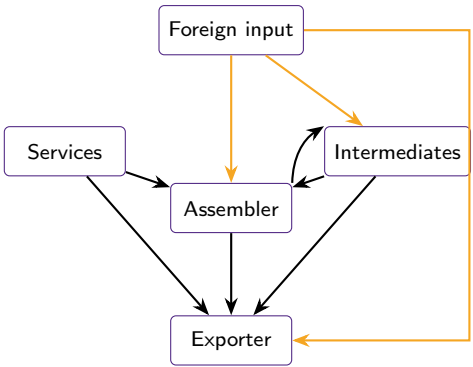


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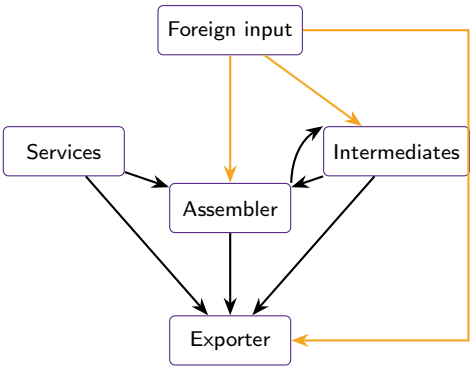


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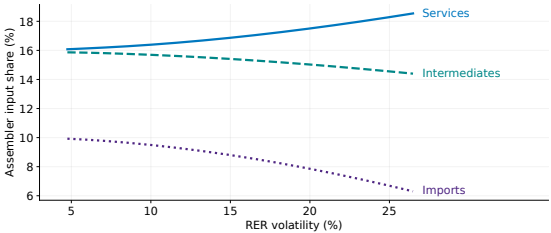
Supplier → buyer; orange arrows denote imports.

- Raise import-price risk: sd(p^*) from 0 to 30%
- Hold means and productivity risk fixed
- Re-solve shares α and prices
- Plot against equilibrium RER volatility

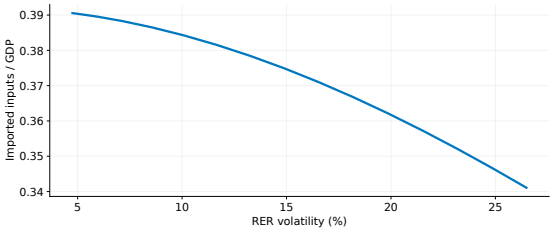
▸ Experiment details ▸ Parametrization ▸ RER volatility

Results: sourcing adjusts and import dependence falls

Domestic sourcing adjusts



Import dependence falls



- Assembler shifts toward services, away from import-intensive suppliers
  - Imported inputs/GDP fall from 39.1% to 34.1%.
- Affects welfare through the level and volatility of consumption.

▸ Welfare

Next steps

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- Counterfactuals: economies facing lower RER volatility
  - Exposure and average output under stabilization
- Calibrate across countries: OECD ICIO and BIS RER data

Thesis plan

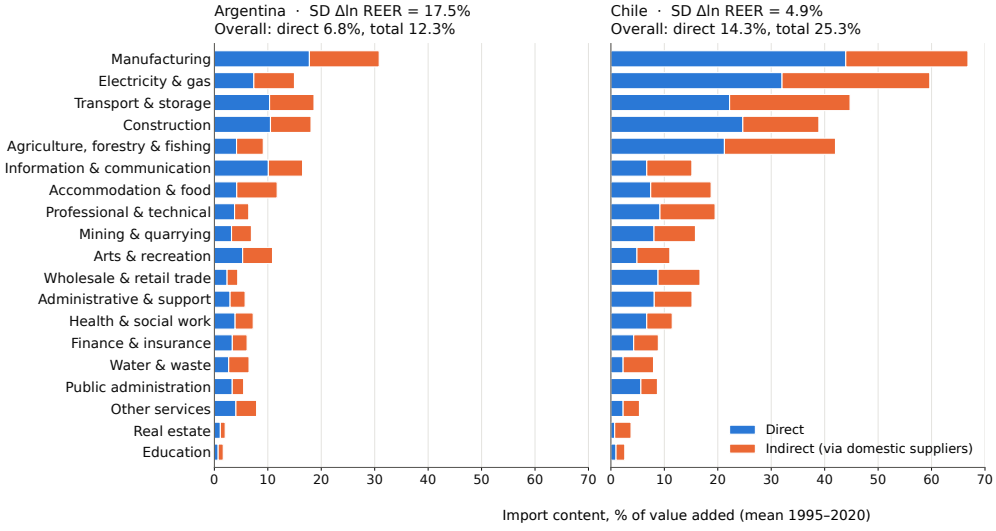
1. **Production networks under real exchange rate uncertainty**
   This project - JMP.

2. **Currency depreciations, credit spreads and imported inputs**
   Summer paper; refine the calibration of impulse responses.

3. **Endogenous production networks and inventory decisions**
   With Vu Cao; inventories hedge future import-price increases but involve storage costs.

Appendix

A0a. Sector import dependence: Argentina and Chile



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A0b. Related literature

1. **Endogenous production networks**
   Oberfield 2018; Acemoglu & Azar 2020; Elliott, Golub & Leduc 2022; Grossman, Helpman & Lhuillier 2023
   - Under uncertainty: Kopytov, Mishra, Nimark & Taschereau-Dumouchel 2024
   - Extend to open economy; exchange rate uncertainty → correlated shocks

2. **Exchange rate propagation**
   Amiti, Itskhoki & Konings 2014, 2019; Auer, Levchenko & Sauré 2019
   - Endogenize the network

Back

A1a. Household problem

The household owns domestic firms and the exporter and supplies $\bar{L}$ inelastically. It chooses state-contingent consumption to maximize

$$\max_{\{C_j, C^*\}} \mathbb{E}[u(C)], \qquad u(C) = \frac{C^{1-\rho} - 1}{1-\rho}, \quad \rho > 0 \quad (u(C) = \ln C \text{ if } \rho = 1).$$

The consumption bundle is Cobb–Douglas:

$$C = \prod_{j:\beta_j > 0} \left(\frac{C_j}{\beta_j}\right)^{\beta_j} \left(\frac{C^*}{\beta^*}\right)^{\beta^*}, \qquad \sum_j \beta_j + \beta^* = 1.$$

Budget constraint,

$$\sum_j P_j C_j + S P^* C^* \leq W\bar{L} + \sum_i \Pi_i + \Pi_X.$$

Competition and constant returns imply zero equilibrium profits, so $Y = W\bar{L}$.

‣ Technology and unit cost ‣ Demand and price index ◂ Environment

A1b. Household demand and the price index

Let μ be the multiplier on the household budget in a given state. The first-order conditions imply

$$u'(C)\frac{\beta_j C}{C_j} = \mu P_j, \qquad u'(C)\frac{\beta^* C}{C^*} = \mu S P^*.$$

Summing expenditures gives $\mu Y = u'(C)C$. Therefore,

$$\boxed{P_j C_j = \beta_j Y, \qquad S P^* C^* = \beta^* Y.}$$

Substitute these demands into the consumption bundle:

$$P = \prod_j P_j^{\beta_j} (S P^*)^{\beta^*}, \qquad \boxed{C = \frac{Y}{P}}, \qquad p = \sum_j \beta_j p_j + \beta^*(s + p^*).$$

The budget binds because utility is increasing in consumption.

Pricing kernel · Household problem

A1c. Household valuation and the sourcing kernel

The marginal utility of domestic-currency income is, up to a common normalization,

$$\Lambda = \mu = \frac{u'(C)}{P} = \frac{C^{-\rho}}{P}, \qquad \ln \Lambda = -\rho \ln C - p.$$

Because $Y = PC$, the valuation weight relevant for sourcing is

$$\boxed{\lambda = \ln(\Lambda Y) = (1-\rho)\ln C.}$$

▸ Continue: technology ◂ Optimal sourcing ◂ Household problem

A2a. Cobb–Douglas technology and cost minimization

For a committed technique, firm i produces with

$$Q_i = e^{z_i} A_i(\alpha_i) \left(\frac{L_i}{\alpha_{iL}}\right)^{\alpha_{iL}} \prod_{j:\alpha_{ij}>0} \left(\frac{X_{ij}}{\alpha_{ij}}\right)^{\alpha_{ij}} \left(\frac{X_{i*}}{\alpha_{i*}}\right)^{\alpha_{i*}}.$$

Shares are nonnegative and sum to one:

$$\alpha_{iL} = 1 - \sum_j \alpha_{ij} - \alpha_{i*}, \qquad a_i(\alpha_i) = \ln A_i(\alpha_i) \text{ is strictly concave.}$$

At realized prices, unit cost solves

$$\mathrm{UC}_i = \min_{L_i, \{X_{ij}\}, X_{i*} \geq 0} \left\{ WL_i + \sum_j P_j X_{ij} + SP^* X_{i*} \right\} \quad \text{subject to } Q_i \geq 1.$$

Let μ_i denote the unit-output multiplier; omit zero-share inputs.

Continue: input demands Environment

A2b. Recovering the unit cost

The unit-output constraint has multiplier μ_i. Input optimality gives

$$L_i = \frac{\mu_i \alpha_{iL}}{W}, \qquad X_{ij} = \frac{\mu_i \alpha_{ij}}{P_j}, \qquad X_{i*} = \frac{\mu_i \alpha_{i*}}{SP^*}.$$

Hence

$$\begin{aligned} \mathrm{UC}_i(\alpha_i, \mathbf{P}, S) &= WL_i + \sum_j P_j X_{ij} + SP^* X_{i*} \\ &= \mu_i \left[\alpha_{iL} + \sum_j \alpha_{ij} + \alpha_{i*} \right] = \mu_i. \end{aligned}$$

Zero-share inputs are omitted.

Continue: unit production Back

A2c. Substitute into unit production

At unit output, the production technology gives

$$1 = e^{z_i} A_i(\alpha_i) \left(\frac{L_i}{\alpha_{iL}}\right)^{\alpha_{iL}} \prod_{j:\alpha_{ij}>0} \left(\frac{X_{ij}}{\alpha_{ij}}\right)^{\alpha_{ij}} \left(\frac{X_{i*}}{\alpha_{i*}}\right)^{\alpha_{i*}}.$$

Substitute the input choices and $\mu_i = \mathrm{UC}_i$. The share terms cancel:

$$1 = e^{z_i} A_i(\alpha_i) \mathrm{UC}_i \, W^{-\alpha_{iL}} \prod_j P_j^{-\alpha_{ij}} (SP^*)^{-\alpha_{i*}}.$$

Thus

$$\mathrm{UC}_i = \frac{W^{\alpha_{iL}} \prod_j P_j^{\alpha_{ij}} (SP^*)^{\alpha_{i*}}}{e^{z_i} A_i(\alpha_i)},$$

$$\boxed{\mathrm{uc}_i = -z_i - a_i(\alpha_i) + \sum_j \alpha_{ij} p_j + \alpha_{i*}(s + p^*) + \alpha_{iL} w.}$$

Continue: balanced trade Back

A2d. Budgets imply balanced trade

1. **Goods clearing:**

$$\sum_j P_j Q_j = \sum_j P_j C_j + \sum_{i,j} P_j X_{ij} + \sum_j P_j X_{Xj}.$$

2. **Add firms' and exporter's zero-profit budgets:**

$$\sum_i P_i Q_i + P_X Q_X = W\bar{L} + \sum_{i,j} P_j X_{ij} + \sum_j P_j X_{Xj} + SP^*\Big(\sum_i X_{i*} + X_{X*}\Big).$$

3. **Cancel domestic transactions and use the household budget:**

$$P_X Q_X = W\bar{L} - \sum_j P_j C_j + SP^*\Big(\sum_i X_{i*} + X_{X*}\Big) = SP^*\Big(C^* + \sum_i X_{i*} + X_{X*}\Big).$$

Back

A3a. Discounted profits and the lognormal identity

For the technique choice, hold state-contingent $\Lambda, Q_i, \mathbf{P}, W, S$ fixed:

$$\max_{\alpha_i} \mathbb{E}[\Lambda Q_i (P_i - \mathrm{UC}_i)] \quad \Longleftrightarrow \quad \min_{\alpha_i} \mathbb{E}[\Lambda Q_i \mathrm{UC}_i].$$

Joint normality gives

$$\mathbb{E}[\Lambda Q_i \mathrm{UC}_i] = \exp\left\{\mathbb{E}[\ln \Lambda + \ln Q_i + \mathrm{uc}_i] + \tfrac{1}{2}\,\mathrm{Var}(\ln \Lambda + \ln Q_i + \mathrm{uc}_i)\right\}.$$

The exponential is increasing. Omitting terms independent of α_i, minimize

$$\boxed{\mathbb{E}[\mathrm{uc}_i] + \tfrac{1}{2}\,\mathrm{Var}(\mathrm{uc}_i) + \mathrm{Cov}(\ln \Lambda, \mathrm{uc}_i) + \mathrm{Cov}(\ln Q_i, \mathrm{uc}_i).}$$

The omitted term is $\mathbb{E}[\ln \Lambda + \ln Q_i] + \tfrac{1}{2}\,\mathrm{Var}(\ln \Lambda + \ln Q_i)$.

Continue: equilibrium step Back

A3b. The equilibrium step

Differentiate first. For an interior choice of α_{ij}:

$$\begin{aligned} 0 &= \mathbb{E}[\partial_{\alpha_{ij}} \mathrm{uc}_i] + \mathrm{Cov}(\mathrm{uc}_i, \partial_{\alpha_{ij}} \mathrm{uc}_i) + \mathrm{Cov}(\ln \Lambda + \ln Q_i, \partial_{\alpha_{ij}} \mathrm{uc}_i) \\ &= \mathbb{E}[\partial_{\alpha_{ij}} \mathrm{uc}_i] + \mathrm{Cov}(\ln \Lambda + \ln Q_i + \mathrm{uc}_i, \partial_{\alpha_{ij}} \mathrm{uc}_i). \end{aligned}$$

Now impose $\mathrm{UC}_i = P_i$ and $P_i Q_i = \omega_i Y$, where ω_i is constant across states:

$$\ln Q_i + \mathrm{uc}_i = \ln \omega_i + \ln Y, \qquad \lambda = \ln(\Lambda Y).$$

Therefore

$$0 = \mathbb{E}[\partial_{\alpha_{ij}} \mathrm{uc}_i] + \mathrm{Cov}(\lambda, \partial_{\alpha_{ij}} \mathrm{uc}_i) = \partial_{\alpha_{ij}} \{\mathbb{E}[\mathrm{uc}_i] + \mathrm{Cov}(\lambda, \mathrm{uc}_i)\}.$$

The lemma gives the same equilibrium first-order conditions, holding λ fixed.

Continue: risk-adjusted prices Back

A3c. Risk-adjusted prices

Rewrite the log unit cost relative to labour:

$$\mathrm{uc}_i = -z_i - a_i(\alpha_i) + w + \sum_j \alpha_{ij}(p_j - w) + \alpha_{i*}(s + p^* - w).$$

Up to terms independent of the firm's technique, the lemma's objective is

$$-a_i(\alpha_i) + \sum_j \alpha_{ij}\mathcal{R}_j + \alpha_{i*}\mathcal{R}_*,$$

where

$$\mathcal{R}_j = \mathbb{E}[p_j - w] + \mathrm{Cov}(\lambda, p_j - w),$$
$$\mathcal{R}_* = \mathbb{E}[s + p^* - w] + \mathrm{Cov}(\lambda, s + p^* - w).$$

For an interior choice,

$$\frac{\partial a_i}{\partial \alpha_{ij}} = \mathcal{R}_j, \qquad \frac{\partial a_i}{\partial \alpha_{i*}} = \mathcal{R}_*.$$

Back

A3d. Extra cost: correlation and volatility

Decompose the extra cost of imported inputs:

$$\text{Cov}(\lambda, s + p^* - w) = \text{Corr}(\lambda, s + p^* - w)\ \text{sd}(\lambda)\ \text{sd}(s + p^* - w)$$

- Higher **nominal exchange rate** or **import price** volatility raises $\text{sd}(s + p^* - w)$
- If depreciations coincide with bad states ($\text{Corr} > 0$), the extra cost rises $\Rightarrow$ lower α_{i*}
- Links the sourcing condition to the empirical figure

Caveat: the correlation is endogenous; it changes with the shares α.

Back

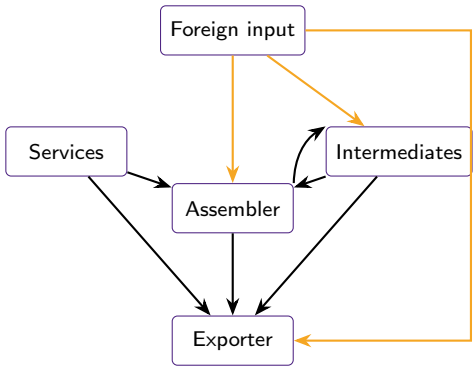
A4a. Impact of a higher risk-adjusted import price

Three domestic sectors and one exporter use labour, domestic inputs and imports.

Numerical experiment: increase $\mathcal{R}_*$

1. Increase foreign import-price uncertainty: sd(p^*) from 0 to 30%.
2. Hold $s = p_X^* = 0$, $\mathbb{E}[p^*] = 0$, and productivity risk fixed.
3. Re-solve all firms' techniques and equilibrium prices at each point.
4. Compare sourcing and import dependence against equilibrium RER volatility.

Benchmark production network



Supplier → buyer; orange arrows denote imports.

▸ Parametrization ▸ RER volatility ◂ Experiment ◂ Results

A4b. Parametrization and normalizations

Rows: assembler, services, intermediates, exporter; columns: suppliers $1, 2, 3, *$.

$$\alpha^o = \begin{pmatrix} 0 & .16 & .16 & .10 \\ .05 & 0 & 0 & 0 \\ .03 & 0 & 0 & .65 \\ .25 & .10 & .15 & .15 \end{pmatrix}, \quad \beta = (.30, .30, .25)^\top, \quad \beta^* = .15, \quad \rho = 4, \quad \bar{L} = 1.$$

$$a_i(\alpha_i) = -\sum_{j \in \mathcal{J}} \kappa_{ij}(\alpha_{ij} - \alpha_{ij}^o)^2 - \kappa_{i0}\left[\sum_{j \in \mathcal{J}}(\alpha_{ij} - \alpha_{ij}^o)\right]^2, \qquad A_i(\alpha_i^o) = 1.$$

$\mathcal{J} = \{1, 2, 3, *\}$; $\kappa_{ij} = (1.5, 6, 6, 3)_i$, $\kappa_{i0} = (3, 6, 6, 6)$; share cap .80.

Prices and exchange rate

$$\mathbb{E}[s] = 0, \quad \sigma_s^2 = 0; \qquad S = 1.$$

$p_X^* = 0$ in every state; $\quad P_X^* = 1.$

$$p^* \sim \mathcal{N}(0, \sigma_{p^*}^2), \quad \sigma_{p^*} \in [0, .30].$$

$\mathbb{E}[p^*] = 0$; $\mathbb{E}[P^*] = e^{\sigma_{p^*}^2/2}$ varies.

Productivity and independence

$$\mathbb{E}[z_i] = 0, \quad \mathrm{sd}(z_i) = .05, \quad i = 1, 2, 3, X.$$

Domestic pairwise correlation: .30.
Export and domestic TFP are independent.
Import-price shocks are independent of TFP.

◂ Example ◂ Experiment

A4c. RER volatility and the experiment

Volatility measure. With $q = s + p^* - p$, the plotted SD (in percent) is

$$\sigma_q = 100\sqrt{\mathrm{Var}(s + p^*) + \mathrm{Var}(p) - 2\,\mathrm{Cov}(s + p^*, p)}.$$

Experiment under A4b's normalizations:

$$s = p_X^* = 0, \quad p^* \sim \mathcal{N}(0, \sigma_{p^*}^2), \quad \sigma_{p^*} \in [0, .30].$$

Under A4b, $s = 0$, so $q = p^* - p$ and

$$\boxed{\sigma_q = 100\sqrt{\sigma_{p^*}^2 + \mathrm{Var}(p) - 2\,\mathrm{Cov}(p^*, p)}}.$$

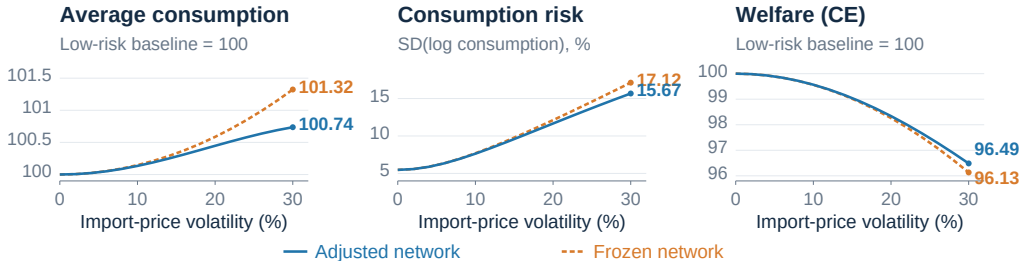
- Keep all other parameters and shock moments in A4b fixed, including $\rho = 4$, $\bar{L} = 1$ and productivity SDs of 5%.
- At each point, re-solve techniques and prices; recompute $\mathrm{Var}(p)$ and $\mathrm{Cov}(p^*, p)$.
- Hold $\mathbb{E}[p^*] = 0$ fixed; the mean price level $\mathbb{E}[P^*] = e^{\sigma_{p^*}^2/2}$ rises.

Log-RER SD across states: 4.75% to 26.48%. Final household imports are excluded from import dependence.

Parametrization Example Results

A4d. Network adjustment: consumption and welfare

Identical import-price shocks; frozen network has α chosen at zero import-price risk; risk aversion $\rho = 4$.



Certainty-equivalent consumption

Defined by $u(C^{CE}) = \mathbb{E}[u(C)]$.

$$C^{CE} = \mathbb{E}[C]\, e^{-\frac{\rho}{2}\,\mathrm{Var}(\log C)}$$

CRRA utility and lognormal consumption.

At 30% import-price volatility

Outcome	Adjustment vs. frozen
Average consumption	−0.58%
Consumption volatility	17.12% → 15.67%
Certainty equivalent	+0.37%

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